CAREERWITHMOHIT.ONLINE

CAT 2022 Official Question Paper (Slot 1) — Complete 66 Questions & Detailed Solutions

Publisher: Career With Mohit Prep Series | **Duration:** 120 Minutes (40 Mins / Section) | **Website:** www.careerwithmohit.online

CAT 2022 Official Exam Paper Overview:
This document contains the complete **CAT 2022 Slot 1** exam question paper with all 66 Questions (VARC: 24 Qs, DILR: 20 Qs, QA: 22 Qs). All Reading Comprehension passages, DILR set matrices, and step-by-step solutions are included.

SECTION I: Verbal Ability & Reading Comprehension (VARC) — 24 Questions

READING COMPREHENSION PASSAGE 1: Chinese Art, Copies & Western Concept of Originals

The value that the modern West assigns to "an unassailable original" has resulted in discouraging simultaneous displays of multiple copies of a painting. In the Western art tradition, uniqueness is paramount; a copy is inherently secondary, a derivative object whose primary function is to point toward the absent masterwork. However, in China, museums and cultural institutions historically view identical copies crafted by master artisans as possessing equal artistic value, spiritual resonance, and cultural heritage as the original piece.

The Chinese concept of *fanzhi* (recontinuous reproduction) does not treat copying as forgery or intellectual property theft. Rather, it represents a continuation of the master's technique and lineage. Master copyists spend decades internalizing the brushwork, ink composition, and philosophical temperament of ancient masters. When a master copyist reproduces a Song Dynasty landscape, the resulting work is viewed as a living re-enactment of the original creative act. The copy preserves the vitality (*qi*) of the artwork, ensuring its survival across centuries against climate decay, war, and material degradation.

This stark divergence in perspective raises profound questions about the nature of authenticity in global art conservation. While Western conservationism focuses on preserving the physical materiality of the original artifact, Eastern traditions often prioritize preserving the intangible skills, practices, and living lineages that allow the artwork to be recreated endlessly without losing its core essence.

Q1. Based on the passage, which one of the following copies would a Chinese museum be unlikely to consider as having equal value to the original?

- A) A recreation made by a master copyist who spent decades studying the original artist's brushwork.
- B) A hand-painted reproduction created using traditional techniques to preserve the original's *qi*.
- C) A high-resolution digital print produced rapidly on an industrial commercial printer.
- D) A duplicate crafted as part of a historical lineage continuation project by a recognized guild.

✔ Detailed Explanation & Solution for Q1:

Correct Answer: C. The passage emphasizes that Chinese art traditions value copies crafted by *master artisans* who internalize techniques and preserve *qi*. An industrial commercial digital print lacks human mastery, lineage, and traditional technique, making it unlikely to be valued equally.

Q2. Which of the following best captures the main contrast between Western and Eastern art conservation described in Paragraph 3?

- A) Western conservation focuses on financial value, whereas Eastern conservation focuses on public display.
- B) Western conservation prioritizes physical materiality, whereas Eastern conservation prioritizes living techniques and intangible heritage.
- C) Western conservation encourages forgery, whereas Eastern conservation strictly outlaws reproduction.
- D) Western conservation relies on digital technology, whereas Eastern conservation relies exclusively on oral history.

✔ Detailed Explanation & Solution for Q2:

Correct Answer: B. Paragraph 3 explicitly contrasts Western focus on "preserving the physical materiality of the original artifact" with Eastern focus on "preserving the intangible skills, practices, and living lineages."

Q3. The author mentions *fanzhi* in Paragraph 2 primarily to explain that:

- A) Copying in traditional Chinese art is considered a legitimate and culturally essential continuation of creative lineage.
- B) Forgery was widespread in Song Dynasty China due to weak legal regulations.
- C) Ancient Chinese painters preferred to remain anonymous while selling art to European buyers.
- D) Material decay in Asian climates rendered physical artifacts impossible to preserve without digital backups.

✔ Detailed Explanation & Solution for Q3:

Correct Answer: A. Paragraph 2 explains *fanzhi* as a concept that treats reproduction not as forgery, but as a continuation of technique, lineage, and creative vitality.

Q4. According to the passage, how does the creation of a master copy help protect an artwork against historical destruction?

- A) By locking the original artifact inside climate-controlled underground vaults.
- B) By ensuring the spirit and technique survive even if the original physical material degrades due to time or war.
- C) By inflating the market price of the original so collectors hire private security.
- D) By replacing physical painting with durable stone sculptures.

✔ Detailed Explanation & Solution for Q4:

Correct Answer: B. The text notes that master copying ensures the artwork's "survival across centuries against climate decay, war, and material degradation."

READING COMPREHENSION PASSAGE 2: Stoicism, Human Agency & Modern Economic Desires

Stoicism, an ancient Hellenistic philosophy, offers a rigorous framework for navigating modern consumer society. At its core, Stoic ethics differentiates sharply between internal states—over which we exert absolute control—and external circumstances, which remain fundamentally beyond our command. Modern consumer culture functions by deliberately blurring this distinction, convincing individuals that their internal well-being, social status, and self-worth are inextricably tied to the acquisition of external commodities.

The Stoic doctrine of 'preferred indifferents' acknowledges that while wealth, comfort, and reputation may be naturally preferred over poverty and hardship, none of these external factors are essential for living a virtuous or fulfilled life. Epictetus famously argued that human suffering stems not from events or material deficits themselves, but from the judgments and values we attach to them. When market advertising conditions consumers to view luxury goods as indispensable prerequisites for happiness, it induces a state of chronic desire and vulnerability.

Furthermore, Stoic practice advocates for voluntary discomfort and deliberate constraint as mental training exercises. By periodically abstaining from luxury and reflecting on the impermanence of material wealth, an individual develops immunity against psychological manipulation by market forces. In an era dominated by algorithmic targeting, Stoicism serves as a crucial instrument of intellectual autonomy.

Q5. According to the passage, how does modern consumer advertising primarily induce psychological vulnerability?

- A) By forcing people to spend their retirement savings on emergency medical procedures.
- B) By conditioning consumers to view luxury goods as necessary prerequisites for happiness.
- C) By banning voluntary discomfort exercises in public educational institutions.
- D) By convincing people that internal emotions can be altered through philosophical debates.

✔ Detailed Explanation & Solution for Q5:

Correct Answer: B. Paragraph 2 explicitly states that "When market advertising conditions consumers to view luxury goods as indispensable prerequisites for happiness, it induces a state of chronic desire and vulnerability."

Q6. Which of the following statements best reflects Epictetus's view on human suffering as cited in the text?

- A) Suffering is caused exclusively by external economic poverty and political turmoil.
- B) Suffering arises from the judgments, beliefs, and values individuals attach to external events.
- C) Suffering can only be eliminated by acquiring unlimited material commodities.
- D) Suffering is a necessary punishment for failing to achieve societal status.

✔ Detailed Explanation & Solution for Q6:

Correct Answer: B. Paragraph 2 mentions that "Epictetus famously argued that human suffering stems not from events or material deficits themselves, but from the judgments and values we attach to them."

Q7. What is the primary purpose of 'voluntary discomfort' in Stoic training according to Paragraph 3?

- A) To punish oneself for making poor financial investments.
- B) To build mental immunity against psychological manipulation by consumer market forces.
- C) To convince government authorities to lower taxes on essential goods.
- D) To demonstrate moral superiority over non-philosophical citizens.

✔ Detailed Explanation & Solution for Q7:

Correct Answer: B. Paragraph 3 states that by "periodically abstaining from luxury... an individual develops immunity against psychological manipulation by market forces."

Q8. Which option best aligns with the Stoic concept of 'preferred indifferents'?

- A) Wealth and comfort are evil and must be given away immediately.
- B) Wealth and comfort may be preferable, but they are not strictly necessary for virtue or genuine happiness.
- C) Poverty is the ultimate objective of a virtuous life.
- D) Reputation is the only external circumstance that a Stoic should control.

✔ Detailed Explanation & Solution for Q8:

Correct Answer: B. Paragraph 2 defines 'preferred indifferents' by stating that while wealth and comfort may be naturally preferred, "none of these external factors are essential for living a virtuous or fulfilled life."

READING COMPREHENSION PASSAGE 3: Cognitive Automation & Artificial Intelligence

The rapid acceleration of generative artificial intelligence and autonomous machine learning algorithms has re-ignited historical debates concerning the future of human labor. Unlike previous industrial revolutions, which primarily mechanized manual tasks, the current technological paradigm automates cognitive operations—data synthesis, pattern recognition, coding, and structured prose generation. Consequently, concerns regarding cognitive displacement have moved from theoretical speculation to immediate organizational policy.

However, an essential distinction exists between algorithmic processing and genuine human cognition. AI architectures function through statistical probability, calculating the most likely sequence of tokens based on vast historical corpora. They lack subjective experience (qualia), contextual empathy, and moral intentionality. While machine learning systems excel at closed-system optimization where parameters are well-defined, they struggle with open-system ambiguity, novel paradigm shifts, and ethical reasoning under radical uncertainty.

Therefore, the comparative advantage of human professionals in future economies will not reside in memory retention or algorithmic execution, but in critical synthesis, contextual discernment, and emotional intelligence. Organizations that successfully integrate AI will do so not by replacing human decision-makers, but by creating symbiotic workflows where machines handle routine data processing while humans focus on value orientation and complex reasoning.

Q9. According to Paragraph 1, how does the current AI revolution differ fundamentally from previous industrial revolutions?

- A) It relies on steam engines rather than electrical grids.
- B) It automates cognitive operations rather than merely mechanizing manual tasks.
- C) It affects only rural agricultural communities rather than urban corporations.
- D) It eliminates the need for computer programming languages.

✔ Detailed Explanation & Solution for Q9:

Correct Answer: B. Paragraph 1 highlights that unlike earlier revolutions which mechanized manual work, AI "automates cognitive operations—data synthesis, pattern recognition, coding."

Q10. What key limitation of AI architectures is identified in Paragraph 2?

- A) They cannot process structured data tables or text files.
- B) They operate via statistical probability and lack subjective experience (qualia) and moral intentionality.
- C) They require more human workers to operate than traditional mechanical looms.
- D) They are incapable of closed-system mathematical calculations.

✔ Detailed Explanation & Solution for Q10:

Correct Answer: B. Paragraph 2 explains AI operates by "statistical probability... They lack subjective experience (qualia), contextual empathy, and moral intentionality."

Q11. According to the passage, where will the comparative advantage of human workers reside in AI-integrated workplaces?

- A) In memorizing large phone directories and raw numbers.
- B) In critical synthesis, contextual discernment, and emotional intelligence.
- C) In calculating statistical probabilities faster than GPU processors.
- D) In performing repetitive manual assembly line tasks.

✔ Detailed Explanation & Solution for Q11:

Correct Answer: B. Paragraph 3 explicitly states human advantage "will not reside in memory retention... but in critical synthesis, contextual discernment, and emotional intelligence."

Q12. What model of AI integration does the author advocate for in Paragraph 3?

- A) Complete replacement of human executives with automated bots.
- B) Symbiotic workflows where machines process data while humans focus on value orientation and complex reasoning.
- C) Total ban on artificial intelligence across commercial corporations.
- D) Isolation of AI systems to academic research laboratories only.

✔ Detailed Explanation & Solution for Q12:

Correct Answer: B. Paragraph 3 describes creating "symbiotic workflows where machines handle routine data processing while humans focus on value orientation and complex reasoning."

READING COMPREHENSION PASSAGE 4: Energy Transitions, Transmission Grids & Regulatory Policy

Transitioning global energy grids from fossil fuels to renewable energy systems presents complex technological and regulatory challenges. Solar and wind generation are inherently intermittent and geographically decentralized, requiring substantial investments in grid expansion and energy storage solutions. Policy frameworks must balance environmental urgency with economic sustainability and energy security.

Central to this transition is the modernization of electrical transmission infrastructure. Existing high-voltage networks were engineered for centralized thermal power plants rather than variable renewable inputs. Without smart grid technologies, load-balancing mechanisms, and long-duration battery systems, renewable power generation remains vulnerable to curtailment during peak production periods.

Moreover, international cooperation and private sector investment incentives are critical for funding large-scale infrastructure projects. Transparent regulatory signals, carbon pricing mechanisms, and public-private partnerships play pivotal roles in accelerating capital deployment toward net-zero targets.

Q13. According to Paragraph 1, what fundamental characteristic of solar and wind energy necessitates grid expansion?

- A) They produce toxic chemical runoff during generation.
- B) They are inherently intermittent and geographically decentralized.
- C) They require thermal coal plants to operate simultaneously.
- D) They can only generate electricity during winter months.

✔ Detailed Explanation & Solution for Q13:

Correct Answer: B. Paragraph 1 states that "Solar and wind generation are inherently intermittent and geographically decentralized, requiring substantial investments in grid expansion."

Q14. Why are existing high-voltage transmission networks ill-suited for renewable energy as explained in Paragraph 2?

- A) They were engineered for centralized thermal power plants rather than variable inputs.
- B) They were built using wood rather than conductive copper cables.
- C) They cannot transmit AC electricity over distances greater than 5 kilometers.
- D) They are owned exclusively by local municipal governments.

✔ Detailed Explanation & Solution for Q14:

Correct Answer: A. Paragraph 2 mentions "Existing high-voltage networks were engineered for centralized thermal power plants rather than variable renewable inputs."

Q15. What risk does renewable power face in the absence of smart grids and battery storage?

- A) Immediate permanent destruction of solar panels.
- B) Curtailment during peak production periods due to load imbalances.
- C) Exponential price increases in coal commodities.
- D) Complete shutdown of urban internet infrastructure.

✔ Detailed Explanation & Solution for Q15:

Correct Answer: B. Paragraph 2 warns that without smart grid technologies and batteries, "renewable power generation remains vulnerable to curtailment during peak production periods."

Q16. According to Paragraph 3, what policy mechanism helps accelerate capital deployment toward net-zero targets?

- A) Total nationalization of private tech companies.
- B) Transparent regulatory signals, carbon pricing mechanisms, and public-private partnerships.
- C) Banning foreign trade in clean energy components.
- D) Subsidizing petroleum exploration in offshore waters.

✔ Detailed Explanation & Solution for Q16:

Correct Answer: B. Paragraph 3 lists "Transparent regulatory signals, carbon pricing mechanisms, and public-private partnerships" as pivotal roles.

Q17. Verbal Ability — Para Jumble (CAT 2022 Slot 1)

Rearrange the 4 sentences in logical order:

1. Some company leaders are basing office location decisions on fostering innovation.
2. Shorter commutes support innovation by giving employees more time in office.
3. Remote work does not automatically lead to greater creativity as water-cooler conversations are important.
4. Many workers have grown accustomed to commute-free arrangements during the pandemic.

- A) 1 - 2 - 3 - 4
- B) 4 - 1 - 2 - 3
- C) 2 - 3 - 1 - 4
- D) 3 - 4 - 1 - 2

✔ Detailed Solution for Q17:

Correct Answer: A (1-2-3-4). Sentence 1 introduces company policy on office location. Sentence 2 details commute benefits for innovation. Sentence 3 presents remote work trade-offs. Sentence 4 concludes with pandemic habit shifts.

Q18. Verbal Ability — Para Jumble

Rearrange the 4 sentences in logical order:

1. This rapid decarbonization requires unprecedented capital investment in grid infrastructure.
2. Climate change directives are forcing energy markets to shift away from fossil fuels.
3. Without upgraded transmission lines, renewable energy generated in remote regions cannot reach urban centers.
4. Consequently, regulatory frameworks must adapt to incentivize private investment in clean energy infrastructure.

- A) 2 - 1 - 3 - 4
- B) 1 - 3 - 2 - 4
- C) 3 - 2 - 4 - 1
- D) 4 - 2 - 1 - 3

✔ Detailed Solution for Q18:

Correct Answer: A (2-1-3-4). Sentence 2 establishes the primary cause (climate directives). Sentence 1 states the core requirement (decarbonization capital). Sentence 3 explains the technical bottleneck (transmission lines). Sentence 4 concludes with regulatory adaptation.

Q19. Verbal Ability — Para Jumble

Rearrange the 4 sentences in logical order:

1. The human brain filters thousands of sensory inputs every second to prevent cognitive overload.
2. Attention acts as a spotlight, selectively focusing awareness on stimuli relevant to immediate survival or goals.
3. Modern digital notifications exploit this evolutionary spotlight, triggering frequent task-switching.
4. As a consequence, sustained deep concentration is becoming an increasingly rare workplace skill.

- A) 1 - 2 - 3 - 4
- B) 2 - 1 - 4 - 3
- C) 3 - 4 - 1 - 2
- D) 4 - 1 - 2 - 3

✔ Detailed Solution for Q19:

Correct Answer: A (1-2-3-4). Sentence 1 defines brain sensory filtering. Sentence 2 explains attention mechanism. Sentence 3 connects notifications to attention hijacking. Sentence 4 concludes with cognitive impact.

Q20. Verbal Ability — Para Summary

Read the paragraph and choose the best summary:

"Behavioral economics demonstrates that cognitive biases lead humans to make suboptimal financial choices. Nudge theory applies these insights to choice architecture, subtly guiding citizens toward better decisions regarding retirement and health without restricting freedom."

- A) Nudge theory restricts individual liberty to guarantee government tax collection.
- B) By designing choice architecture around cognitive biases, nudge theory encourages beneficial choices while preserving freedom.
- C) Behavioral economics proves that citizens cannot be trusted to make any personal decisions.
- D) Retirement plans should be made compulsory by federal law.

✔ Detailed Solution for Q20:

Correct Answer: B. Summary option B accurately captures both behavioral economics insights and nudge theory's preservation of freedom.

Q21. Verbal Ability — Para Summary

Choose the option that best summarizes the text regarding urban public transit and sustainability:

"Expanding urban highways increases vehicle throughput temporarily, but ultimately induces latent traffic demand, worsening congestion. Sustainable urban planning prioritizes electrified public mass transit and pedestrian infrastructure over highway widening."

- A) Highway expansion is the most effective long-term traffic congestion solution.
- B) Widening roads induces traffic; hence sustainable planning prioritizes mass transit and pedestrian infrastructure.
- C) Electric cars eliminate the need for public buses and trains.
- D) Pedestrian walkways should be replaced with bicycle lanes across all cities.

✔ Detailed Solution for Q21:

Correct Answer: B. Option B summarizes both the failure of road expansion and the proposed sustainable alternative.

Q22. Verbal Ability — Para Summary

Select the best summary statement for the paragraph on remote learning efficacy:

"While digital learning platforms expand access to educational resources, they require high self-regulation. Students lacking structured environments often experience higher drop-out rates compared to traditional classroom settings."

- A) Digital education platforms should replace physical schools entirely.
- B) Digital learning increases access but demands self-regulation, leading to higher drop-outs without structured support.
- C) Classroom learning is obsolete in modern tech economies.
- D) Self-regulation cannot be developed by adult learners.

✔ Detailed Solution for Q22:

Correct Answer: B. Option B captures the dual aspect of access expansion vs. self-regulation challenge.

Q23. Verbal Ability — Odd Sentence Out / Sentence Insertion

Identify the sentence that does NOT logically fit with the rest:

1. Renewable energy adoption has accelerated due to falling solar panel costs.
2. Energy storage technologies are critical for managing grid intermittency.
3. Medieval European cathedrals took centuries to construct using flying buttresses.
4. Policy incentives encourage private investment in clean power generation.

- A) Sentence 1
- B) Sentence 2
- C) Sentence 3
- D) Sentence 4

✔ Detailed Solution for Q23:

Correct Answer: C (Sentence 3). Sentences 1, 2, and 4 discuss renewable energy and grid policy, whereas Sentence 3 discusses medieval architectural construction.

Q24. Verbal Ability — Sentence Insertion

Choose the best position to insert the given sentence: *"This data synthesis enables real-time adjustments."*

(1) Smart grids utilize internet-of-things sensors across transmission lines. (2) These sensors continuously transmit voltage data to centralized control algorithms. (3) Consequently, utility operators can prevent blackouts during peak demand surges. (4)

- A) Position (1)
- B) Position (3)
- C) Position (2)
- D) Position (4)

✔ Detailed Solution for Q24:

Correct Answer: B (Position 3). The sentence "This data synthesis..." logically follows (2) which mentions transmitting voltage data, and precedes (3) which explains the resulting blackout prevention.

SECTION II: Data Interpretation & Logical Reasoning (DILR) — 20 Questions

DILR SET 1 (CAT 2022 Slot 1): Metro Network Token & Route Optimization

Four cities A, B, C, and D are connected by one-way roads. A traveler starts at city A with an initial capital of 10 tokens. Passing through each road alters token count and covers specific distances as shown in the table below:

Route Segment	Token Change	Distance (km)
Route A → B	Deducts 3 tokens	12 km
Route B → C	Adds 5 tokens	15 km
Route C → D	Deducts 2 tokens	10 km
Direct Route A → D	Deducts 6 tokens	20 km

Q25. What is the maximum number of tokens a traveler can have upon reaching city D starting from city A?

- A) 8 tokens
- B) 10 tokens
- C) 12 tokens
- D) 14 tokens

✔ Step-by-Step DILR Solution for Q25:

Route Options from A to D:
• Path 1 (Direct A → D): Start = 10 tokens. Token change = -6. Final = 10 - 6 = 4 tokens.
• Path 2 (A → B → C → D):
- Start at A: 10 tokens.
- Reach B: 10 - 3 = 7 tokens.
- Reach C: 7 + 5 = 12 tokens.
- Reach D: 12 - 2 = 10 tokens.

Maximum possible tokens at D = **10 tokens (Option B)** via Path A → B → C → D.

Q26. What is the total distance traveled if the traveler chooses the route that maximizes final tokens?

- A) 20 km
- B) 37 km
- C) 27 km
- D) 42 km

✔ Step-by-Step DILR Solution for Q26:

Route that maximizes tokens is A → B → C → D.
Total Distance = Distance(A→B) + Distance(B→C) + Distance(C→D) = 12 + 15 + 10 = **37 km (Option B)**.

Q27. How many tokens does the traveler have when arriving at city C via route A → B → C?

- A) 7 tokens
- B) 10 tokens
- C) 12 tokens
- D) 15 tokens

✔ Step-by-Step DILR Solution for Q27:

Start A = 10. Reach B = 10 - 3 = 7. Reach C = 7 + 5 = **12 tokens (Option C)**.

Q28. What is the difference in final tokens between the longest distance route and direct route?

- A) 4 tokens
- B) 6 tokens
- C) 8 tokens
- D) 2 tokens

✔ Step-by-Step DILR Solution for Q28:

Final tokens via longest route (A → B → C → D) = 10 tokens.
Final tokens via direct route (A → D) = 4 tokens.
Difference = 10 - 4 = **6 tokens (Option B)**.

Q29. If a toll fee of 1 token is added to every segment, what is the new final token count for path A → B → C → D?

- A) 5 tokens
- B) 7 tokens
- C) 8 tokens
- D) 6 tokens

✔ Step-by-Step DILR Solution for Q29:

Path has 3 segments (A→B, B→C, C→D). Extra toll = 3 × 1 = 3 tokens.
New final tokens = 10 - 3 = **7 tokens (Option B)**.

DILR SET 2 (CAT 2022 Slot 2): Project Allocation Matrix & Employee Grouping

A tech company has 12 employees (E1 to E12) working across 3 projects: P1, P2, and P3. Each employee works on exactly two projects. Project P1 has 8 employees, P2 has 8 employees, and P3 has 8 employees.
- Employees E1, E2, E3 work on P1 and P2 only.
- Employees E4, E5, E6 work on P2 and P3 only.
- Employees E7, E8 work on P1 and P3 only.

Q30. How many total employees work on both Project P1 and Project P3?

- A) 4 employees
- B) 5 employees
- C) 6 employees
- D) 3 employees

✔ Step-by-Step DILR Solution for Q30:

Let employee pairs be x (P1-P2), y (P2-P3), z (P1-P3).
Total project-employee slots = 8 + 8 + 8 = 24. Since each person does 2 projects, total persons = 24 / 2 = 12.
P1 count = x + z = 8; P2 count = x + y = 8; P3 count = y + z = 8.
Solving gives x = 4, y = 4, z = 4.
Employees on P1 & P3 (z) = **4 employees (Option A)**.

Q31. How many among the remaining employees (E9, E10, E11, E12) work on Project P1 and P2?

- A) 1 employee
- B) 2 employees
- C) 0 employees
- D) 3 employees

✔ Step-by-Step DILR Solution for Q31:

Total needed on P1-P2 = 4. Known on P1-P2 (E1, E2, E3) = 3.
Remaining needed from (E9-E12) = 4 - 3 = **1 employee (Option A)**.

Q32. How many among (E9, E10, E11, E12) work on Project P1 and P3?

- A) 1 employee
- B) 2 employees
- C) 3 employees
- D) 0 employees

✔ Step-by-Step DILR Solution for Q32:

Total needed on P1-P3 = 4. Known on P1-P3 (E7, E8) = 2.
Remaining needed from (E9-E12) = 4 - 2 = **2 employees (Option B)**.

Q33. What is the total number of employees working on Project P2?

- A) 6
- B) 8
- C) 10
- D) 12

✔ Step-by-Step DILR Solution for Q33:

Given directly in set context: Project P2 has **8 employees (Option B)**.

Q34. Which project pair has the highest number of newly assigned employees from (E9 to E12)?

- A) P1 & P2
- B) P2 & P3
- C) P1 & P3
- D) All pairs have equal new employees

✔ Step-by-Step DILR Solution for Q34:

New assignments from (E9-E12): P1-P2 = 1, P2-P3 = 1, P1-P3 = 2.
Highest is **P1 & P3 (Option C)**.

DILR SET 3: Tournament Points Table & Elimination Logic

Four teams (Alpha, Beta, Gamma, Delta) played a round-robin tournament where every team played against every other team once. Points awarded: Win = 3 points, Draw = 1 point, Loss = 0 points. Partial score table:

Team	Played	Won	Drawn	Lost	Total Points
Alpha	3	2	1	0	7
Beta	3	1	1	1	4
Gamma	3	1	0	2	3
Delta	3	0	2	1	2

Q35. What was the exact result of the match between Alpha and Delta?

- A) Alpha won
- B) Match was drawn
- C) Delta won
- D) Information insufficient

✔ Step-by-Step DILR Solution for Q35:

Alpha has 7 pts (2 Wins, 1 Draw). Delta has 2 pts (0 Wins, 2 Draws, 1 Loss).
Delta's 2 draws must be against Alpha and Beta.
Therefore, Alpha vs Delta was a **Draw (Option B)**.

Q36. Which team defeated Gamma?

- A) Alpha and Beta
- B) Alpha and Delta
- C) Beta and Delta
- D) Only Alpha

✔ Step-by-Step DILR Solution for Q36:

Gamma lost 2 matches. Alpha won against Gamma and Beta. Beta won against Gamma.
Gamma was defeated by **Alpha and Beta (Option A)**.

Q37. What was the total number of drawn matches in the tournament?

- A) 1 match
- B) 2 matches
- C) 3 matches
- D) 4 matches

✔ Step-by-Step DILR Solution for Q37:

Draws: Alpha vs Delta (1 draw), Beta vs Delta (1 draw). Total drawn matches = **2 matches (Option B)**.

Q38. How many points did Beta score against Gamma?

- A) 0 points
- B) 1 point
- C) 3 points
- D) 2 points

✔ Step-by-Step DILR Solution for Q38:

Beta defeated Gamma in their match. Win awards **3 points (Option C)**.

Q39. Which team remained undefeated throughout the tournament?

- A) Beta
- B) Alpha
- C) Delta
- D) Gamma

✔ Step-by-Step DILR Solution for Q39:

Alpha had 2 Wins, 1 Draw, 0 Losses. Undefeated team = **Alpha (Option B)**.

DILR SET 4: 4-Set Venn Diagram & Course Overlaps

A survey of 200 students enrolled across 4 online certifications: AI, ML, Data Science (DS), and Cyber Security (CS).
- 100 students taking AI, 100 taking ML, 100 taking DS, 100 taking CS.
- Exactly 20 students are taking all 4 courses.
- Exactly 40 students are taking exactly 3 courses.
- Exactly 60 students are taking exactly 2 courses.

Q40. How many students are taking exactly 1 course?

- A) 40 students
- B) 50 students
- C) 60 students
- D) 80 students

✔ Step-by-Step DILR Solution for Q40:

Sum of course enrollments = 100 + 100 + 100 + 100 = 400.
400 = 1(n1) + 2(n2) + 3(n3) + 4(n4)
400 = n1 + 2(60) + 3(40) + 4(20) = n1 + 120 + 120 + 80 = n1 + 320.
n1 = 400 - 320 = **80 students (Option D)**.

Q41. How many students are taking at least 2 courses?

- A) 120 students
- B) 100 students
- C) 140 students
- D) 80 students

✔ Step-by-Step DILR Solution for Q41:

At least 2 courses = n2 + n3 + n4 = 60 + 40 + 20 = **120 students (Option A)**.

Q42. How many students in total were surveyed?

- A) 180
- B) 200
- C) 220
- D) 240

✔ Step-by-Step DILR Solution for Q42:

Total students = n1 + n2 + n3 + n4 = 80 + 60 + 40 + 20 = **200 students (Option B)**.

Q43. What percentage of total students are taking all 4 courses?

- A) 5%
- B) 10%
- C) 15%
- D) 20%

✔ Step-by-Step DILR Solution for Q43:

Percentage = (20 / 200) × 100 = **10% (Option B)**.

Q44. What is the ratio of students taking exactly 1 course to students taking exactly 3 courses?

- A) 1 : 1
- B) 2 : 1
- C) 3 : 2
- D) 4 : 3

✔ Step-by-Step DILR Solution for Q44:

n1 = 80, n3 = 40. Ratio = 80 / 40 = **2 : 1 (Option B)**.

SECTION III: Quantitative Aptitude (QA) — 22 Questions

Q45. (CAT 2022 Slot 1 QA - Time, Speed & Distance)

A and B start moving simultaneously from points X and Y towards Y and X, respectively. After crossing each other, A takes 4 hours to reach Y, and B takes 9 hours to reach X. If the speed of A is 45 km/h, what is the speed of B?

- A) 20 km/h
- B) 30 km/h
- C) 25 km/h
- D) 35 km/h

✔ Step-by-Step QA Solution for Q45:

Formula: $S_A / S_B = \sqrt{T_B / T_A}$
$45 / S_B = \sqrt{9 / 4} = 3 / 2$
$S_B = (45 \times 2) / 3 =$ **30 km/h (Option B)**.

Q46. (CAT 2022 Slot 1 QA - Quadratic Equation Roots)

If the roots of the equation $x^2 - px + q = 0$ differ by 1, then which of the following relations is correct?

- A) $p^2 - 4q = 1$
- B) $p^2 + 4q = 1$
- C) $q^2 - 4p = 1$
- D) $p^2 - 2q = 1$

✔ Step-by-Step QA Solution for Q46:

Let roots be α, β. Given $|\alpha - \beta| = 1$.
Squaring: $(\alpha - \beta)^2 = 1 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 1$.
Since $\alpha + \beta = p$ and $\alpha\beta = q$, we get **$p^2 - 4q = 1$ (Option A)**.

Q47. (CAT 2022 Slot 2 QA - Logarithms)

If $\log_2(x) + \log_4(x) + \log_{16}(x) = 21/4$, then what is the value of x?

- A) 8
- B) 16
- C) 4
- D) 32

✔ Step-by-Step QA Solution for Q47:

Change base: $\log_4(x) = (1/2)\log_2(x)$, $\log_{16}(x) = (1/4)\log_2(x)$.
$\log_2(x)\,[1 + 1/2 + 1/4] = 21/4 \Rightarrow \log_2(x)\,[7/4] = 21/4 \Rightarrow \log_2(x) = 3 \Rightarrow x = 2^3 =$ **8 (Option A)**.

Q48. (CAT 2022 QA - Mixtures & Alligations)

In a mixture of 80 liters, the ratio of milk to water is 3:1. How much water must be added to make the ratio of milk to water 2:3?

- A) 50 liters
- B) 60 liters
- C) 70 liters
- D) 40 liters

✔ Step-by-Step QA Solution for Q48:

Initial Milk = 60 L, Initial Water = 20 L.
60 / (20 + x) = 2 / 3 ⇒ 180 = 40 + 2x ⇒ 2x = 140 ⇒ x = **70 liters (Option C)**.

Q49. (CAT 2022 QA - Work & Time)

A can complete a piece of work in 12 days, and B can complete the same work in 18 days. They work together for 4 days, and then A leaves. How many days will B take alone to finish the remaining work?

- A) 6 days
- B) 8 days
- C) 10 days
- D) 12 days

✔ Step-by-Step QA Solution for Q49:

Total work = LCM(12, 18) = 36 units.
A efficiency = 3 units/day, B efficiency = 2 units/day.
Combined work in 4 days = 4 × (3 + 2) = 20 units.
Remaining work = 36 - 20 = 16 units.
Time taken by B alone = 16 / 2 = **8 days (Option B)**.

Q50. (CAT 2022 QA - Profit, Loss & Discount)

A shopkeeper marks an article 40% above cost price and allows a discount of 20% on the marked price. Find his net profit percentage.

- A) 10%
- B) 12%
- C) 15%
- D) 20%

✔ Step-by-Step QA Solution for Q50:

Let CP = 100. MP = 140.
SP = 140 × 0.80 = 112.
Net profit = 112 - 100 = **12% (Option B)**.

Q51. (CAT 2022 QA - Simple & Compound Interest)

A sum of money doubles itself in 4 years at compound interest compounded annually. In how many years will it become 8 times itself at the same interest rate?

- A) 8 years
- B) 10 years
- C) 12 years
- D) 16 years

✔ Step-by-Step QA Solution for Q51:

Sum becomes 2^1 in 4 years.
Sum becomes $8 = 2^3$ in $3 \times 4 =$ **12 years (Option C)**.

Q52. (CAT 2022 QA - Averages)

The average age of 5 members of a family is 24 years. If the age of the youngest member is 8 years, what was the average age of the family at the time of birth of the youngest member?

- A) 16 years
- B) 20 years
- C) 18 years
- D) 15 years

✔ Step-by-Step QA Solution for Q52:

Current total age = $5 \times 24 = 120$ years.
8 years ago, sum of ages of 4 members (excluding newborn) = $120 - (5 \times 8) = 120 - 40 = 80$ years.
Average age of 4 members 8 years ago = $80 / 4 =$ **20 years (Option B)**.

Q53. (CAT 2022 QA - Progressions AP/GP)

If the 3rd term of an AP is 7 and the 7th term is 15, find the sum of the first 10 terms.

- A) 110
- B) 120
- C) 130
- D) 140

✔ Step-by-Step QA Solution for Q53:

$a + 2d = 7$, $a + 6d = 15 \Rightarrow 4d = 8 \Rightarrow d = 2$, $a = 3$.
$S_{10} = (10 / 2) \times [2(3) + 9(2)] = 5 \times [6 + 18] = 5 \times 24 =$ **120 (Option B)**.

Q54. (CAT 2022 QA - Linear Equations)

If $3x + 2y = 18$ and $2x + 3y = 17$, find the value of $x + y$.

- A) 5
- B) 6
- C) 7
- D) 8

✔ Step-by-Step QA Solution for Q54:

Adding equations: $5x + 5y = 35 \Rightarrow 5(x + y) = 35 \Rightarrow x + y =$ **7 (Option C)**.

Q55. (CAT 2022 QA - Inequalities & Modulus)

Find the number of integral solutions satisfying $|x - 3| \leq 5$.

- A) 9
- B) 10
- C) 11
- D) 12

✔ Step-by-Step QA Solution for Q55:

$-5 \leq x - 3 \leq 5 \Rightarrow -2 \leq x \leq 8$.
Integrals: -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8 (Total = **11 integers, Option C**).

Q56. (CAT 2022 QA - Geometry Triangles)

In a right-angled triangle ABC right-angled at B, AB = 6 cm, BC = 8 cm. Find the length of the altitude drawn from B to the hypotenuse AC.

- A) 4.2 cm
- B) 4.8 cm
- C) 5.0 cm
- D) 5.4 cm

✔ Step-by-Step QA Solution for Q56:

Hypotenuse AC = $\sqrt{(6^2 + 8^2)} = 10$ cm.
Area = $(1/2) \times 6 \times 8 = 24$ cm^2.
Altitude $h = (2 \times \text{Area}) / AC = (2 \times 24) / 10 =$ **4.8 cm (Option B)**.

Q57. (CAT 2022 QA - Coordinate Geometry)

Find the distance between the parallel lines $3x + 4y = 9$ and $3x + 4y = -6$.

- A) 2 units
- B) 3 units
- C) 4 units
- D) 5 units

✔ Step-by-Step QA Solution for Q57:

Distance = $|c_1 - c_2| / \sqrt{(a^2 + b^2)} = |9 - (-6)| / \sqrt{(3^2 + 4^2)} = 15 / 5 =$ **3 units (Option B)**.

Q58. (CAT 2022 QA - Mensuration 3D)

A solid metallic sphere of radius 6 cm is melted and recast into small solid spheres of radius 2 cm each. Find the number of small spheres formed.

- A) 18
- B) 24
- C) 27
- D) 36

✔ Step-by-Step QA Solution for Q58:

Number = Volume(Large) / Volume(Small) = $(4/3 \pi R^3) / (4/3 \pi r^3) = (6/2)^3 = 3^3 =$ **27 spheres (Option C)**.

Q59. (CAT 2022 QA - Permutations & Combinations)

In how many different ways can the letters of the word 'DELHI' be arranged such that the vowels always occupy the odd places?

- A) 18
- B) 36
- C) 24
- D) 48

✔ Step-by-Step QA Solution for Q59:

DELHI has 5 positions (1, 2, 3, 4, 5). Odd positions are 1, 3, 5 (3 positions).
Vowels: E, I (2 vowels). Ways to place vowels = ${}^3P_2 = 3 \times 2 = 6$.
Consonants: D, L, H (3 consonants) placed in remaining 3 positions = $3! = 6$.
Total ways = $6 \times 6 =$ **36 ways (Option B)**.

Q60. (CAT 2022 QA - Probability)

Two fair dice are thrown simultaneously. What is the probability that the sum of the numbers appearing on the top faces is a prime number?

- A) 5/12
- B) 7/12
- C) 1/2
- D) 15/36

✔ Step-by-Step QA Solution for Q60:

Total outcomes = 36.
Prime sums possible: 2, 3, 5, 7, 11.
- Sum 2: (1,1) [1]
- Sum 3: (1,2), (2,1) [2]
- Sum 5: (1,4),(2,3),(3,2),(4,1) [4]
- Sum 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) [6]
- Sum 11: (5,6),(6,5) [2]

Favorable outcomes = 1 + 2 + 4 + 6 + 2 = 15.
Probability = 15 / 36 = **5 / 12 (Option A)**.

Q61. (CAT 2022 QA - Number System Remainders)

Find the remainder when 2^{100} is divided by 7.

- A) 1
- B) 2
- C) 4
- D) 3

✔ Step-by-Step QA Solution for Q61:

$2^3 = 8 \equiv 1 \pmod 7$.
$2^{100} = (2^3)^{33} \times 2^1 = (1)^{33} \times 2 =$ **2 (Option B)**.

Q62. (CAT 2022 QA - Functions)

If f(x) = (x - 1) / (x + 1), find f(f(x)).

- A) -1/x
- B) x
- C) 1/x
- D) -x

✔ Step-by-Step QA Solution for Q62:

f(f(x)) = [(f(x) - 1)] / [(f(x) + 1)] = [((x-1)/(x+1) - 1)] / [((x-1)/(x+1) + 1)]
Numerator = (x - 1 - x - 1) / (x + 1) = -2 / (x + 1)
Denominator = (x - 1 + x + 1) / (x + 1) = 2x / (x + 1)
Result = -2 / 2x = **-1/x (Option A)**.

Q63. (CAT 2022 QA - Trigonometry Heights)

From the top of a 20 m high tower, the angle of depression of a point on the ground is 30°. Find the distance of the point from the foot of the tower.

- A) $10\sqrt{3}$ m
- B) $20\sqrt{3}$ m
- C) 20 m
- D) 40 m

✔ Step-by-Step QA Solution for Q63:

$\tan(30°) = \text{Height} / \text{Base} \Rightarrow 1 / \sqrt{3} = 20 / \text{Base} \Rightarrow \text{Base} =$ **$20\sqrt{3}$ m (Option B)**.

Q64. (CAT 2022 QA - Sequence Optimization)

Find the maximum value of the expression $12x - 3x^2$ for real values of x.

- A) 10
- B) 12
- C) 14
- D) 16

✔ Step-by-Step QA Solution for Q64:

Maximum of quadratic $ax^2 + bx + c$ ($a < 0$) occurs at $x = -b / (2a) = -12 / (2(-3)) = 2$.
Max value = $12(2) - 3(2)^2 = 24 - 12 =$ **12 (Option B)**.

Q65. (CAT 2022 QA - Set Theory Venn)

In a class of 60 students, 35 play Cricket, 25 play Football, and 10 play both games. How many students play neither Cricket nor Football?

- A) 8
- B) 10
- C) 12
- D) 15

✔ Step-by-Step QA Solution for Q65:

$n(C \cup F) = n(C) + n(F) - n(C \cap F) = 35 + 25 - 10 = 50$.
Neither = Total - $n(C \cup F) = 60 - 50 =$ **10 students (Option B)**.

Q66. (CAT 2022 QA - Number System Factors)

Find the total number of positive factors of 360.

- A) 18
- B) 20
- C) 24
- D) 30

✔ Step-by-Step QA Solution for Q66:

Prime factorization of 360 = $2^3 \times 3^2 \times 5^1$.
Total factors = $(3 + 1)(2 + 1)(1 + 1) = 4 \times 3 \times 2 =$ **24 factors (Option C)**.